

Sine and cosine functions



1 Differentiate the following functions:

a $y = \cos 3x$

c $y = 2 - 3 \cos x$

e $y = -3 \sin x$

g $y = \sin (4x^5 + 3)$

i $y = \sin (x^4 + 3)$

k $y = \sin \left(\frac{x}{3} \right) - 2 \cos x$

m $y = 4 \sin \left(\frac{t}{4} \right) + 3 \cos 4t + t^4$

b $y = \sin 7x$

d $y = \sin x + 3 \cos x$

f $y = \sin (x^6 + 5)$

h $y = 3x + \sin 5x$

j $y = \cos (3x^2)$

l $y = \sin x^5$

2 Differentiate the following functions:

a $y = 4x \cos x$

c $y = \sin^6 (3x)$

e $y = \sqrt{\cos 4x}$

g $y = \sin^2 (5x)$

i $y = (x^4 + 3) \cos 5x$

k $y = x^{\frac{7}{4}} \cos 2x$

m $y = \frac{3x}{\sin x}$

o $y = x^3 \cos x$

b $y = \sin x \cos x$

d $y = \sin 4x (2 + \cos x)$

f $y = \cos^2 \left(x + \frac{\pi}{2} \right)$

h $y = \frac{\sin x}{x - 8}$

j $y = \cos 4x \sin 6x$

l $y = 3 \cos 4x + 6 \sin 5x$

n $y = x^4 \sin \left(\frac{x}{2} \right)$

p $y = 4 \sin 5x \cos 5x$

Applications

3 Differentiate the function $xy = \sin 5x$.

4 Consider $\tan x = \frac{\sin x}{\cos x}$ and differentiate $y = \tan x$ using the quotient rule.

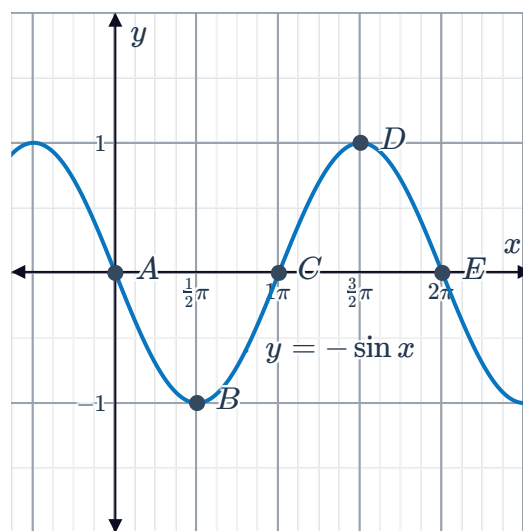
5 Explain why the gradient of the function $y = \sin^2 (5x) + \cos^2 (5x)$ equal to 0 for all x .

- 6 Consider the graph of $y = -\sin x$ which is the gradient function of $y = \cos x$. A number of points have been labelled on the graph.



Name the point on the gradient function that corresponds to the following locations on the graph of $y = \cos x$:

- Where $y = \cos x$ is increasing most rapidly.
- Where $y = \cos x$ is decreasing most rapidly.
- Where $y = \cos x$ is stationary.



- 7 There is an expansion system in mathematics that allows a function to be written in terms of powers of x . The value of $\sin x$ and $\cos x$, for any value of x , can be given by the expansions below:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$

Use the expansions to find:

- $\frac{d}{dx}(\sin x)$ in terms of x .
- $\frac{d}{dx}(\cos x)$ in terms of x .
- Hence express the derivatives of $\sin x$ and $\cos x$ in simplest form.